

ENGINEERING DESIGN PORTFOLIO



CHIMWEMWE CHINKUYU

B.S.E. Mechanical & Aerospace Engineering · Minor in Computer Science · Princeton University, Class of 2026

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EDUCATION

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I'm inspired by the possibility to turn ideas into smart, functional solutions at the intersection of hardware and software. My work spans mechanical design, fabrication, and software development — from building a load-bearing airfoil to writing autonomous vehicle controllers in Python and ROS.

I'm eager to grow in innovation-focused environments and contribute to advanced devices, autonomous systems, and technologies that enhance human health and quality of life.



TECHNICAL SKILLS

CAD / CAM

- PTC Creo (incl. FEA)
- Fusion 360
- Onshape
- AutoCAD

Programming & Robotics

- Python
- MATLAB
- Java
- C
- ROS
- COMPAS FAB
- UR-RTDE
- PID control

Fabrication

- 3D Printing (FDM, SLA, SLS)
- Manual & CNC Machining
- Carpentry
- Robotic Fabrication
- Autonomous Assembly

Software & Tools

- Simulink
- Jupyter Notebook
- Docker
- Anaconda
- Grasshopper
- NVIDIA Jetson
- RadiAnt DICOM Viewer

LINKEDIN

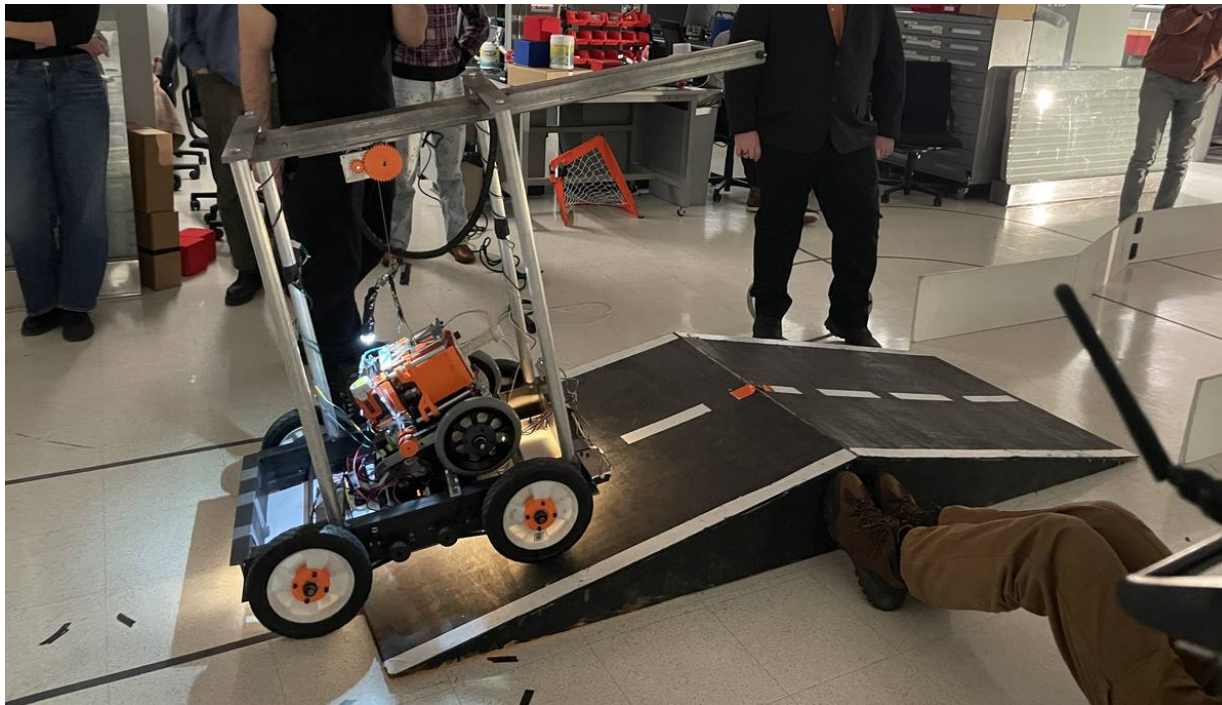
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PORTFOLIO WEBSITE

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MECHANICAL ENGINEERING

Design & Manufacturing of a Search and Rescue Robot

Oct – Dec 2025

OUTCOME

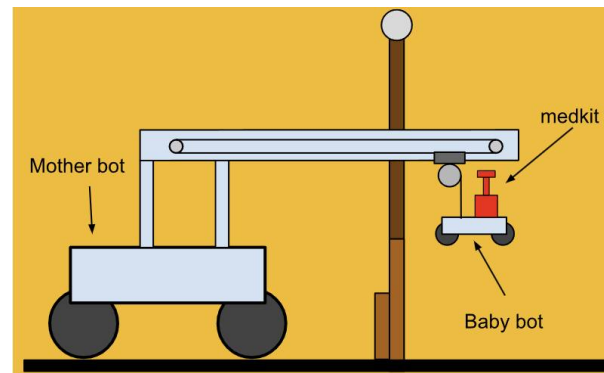
An 11-member team built a marsupial search-and-rescue robot in 8 weeks. The Motherbot autonomously navigated a ramp, chute, and 36-inch wall while carrying the Babybot. The Babybot was deployed over the wall, navigated to a target zone, and delivered a medkit payload — completing the full course at the final class demonstration.

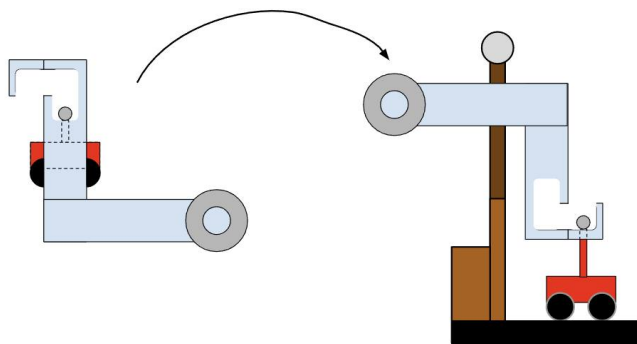
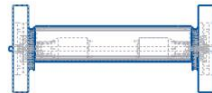
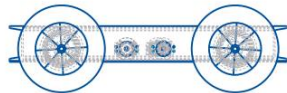
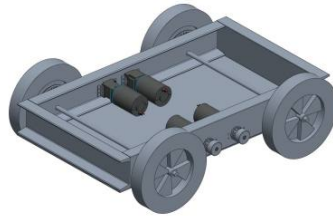
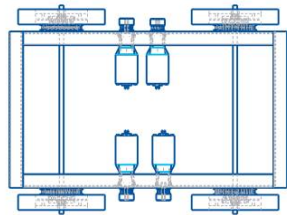
MY CONTRIBUTION

I designed the Motherbot's low-carbon steel (AISI 1018) chassis and a 60:1 two-stage HTD belt drivetrain. Work included FEA for structural verification, manufacturing drawings in Creo Parametric, and hands-on fabrication: band saw cutting, knee mill coping, and MIG welding.

SKILLS USED

Creo Parametric · FEA · GD&T · MIG Welding · CNC Milling · Lathe · Belt Drive Design · Embedded Control (Teensy/C++)





MECHANICAL ENGINEERING

Search and Rescue Robot — Subsystem Detail

Oct – Dec 2025

CHASSIS DESIGN

The welded steel chassis was optimized for rigidity at motor mount positions and the lifting arm pivot. FBD analysis confirmed tip-over stability during Babybot deployment. The weldment showed no deformation through all testing.

DRIVETRAIN

60:1 two-stage belt reduction over HTD pulleys and a jackshaft. Belt drives chosen for shock absorption and tolerance of minor shaft misalignment. Adjustment slots in motor mount brackets allowed precise pulley alignment.

LIFTING ARM

Rotating arm pivots from stowed horizontal to fully raised, clearing the wall. A spring-latch cradle keeps the Babybot level during lift and releases on contact with the far side. Torque requirements calculated analytically.

CONTROLS

State machine: proportional control (flat) → bang-bang (ramp) → PD (chute) → open-loop arm sequence (wall). Ultrasonic sensors trigger state transitions; CdS phototransistors provide line-following error signals.

Custom Differential Gearbox

June 2025

OUTCOME

A fully parametric bevel-gear axle differential designed entirely in Autodesk Fusion 360. The planetary differential configuration distributes drive torque between two output shafts while allowing independent rotation speeds, enabling smooth, skid-free turning.

DESIGN APPROACH

Gear geometry was determined using standard bevel gear formulas driven by linked Fusion 360 equations — module, pressure angle, tooth count, and pitch circle diameter update the full geometry automatically as inputs change.

COMPONENTS MODELED

Bevel ring gear · Spider gears · Output half-shafts · Hub flanges · Outer casing. All parts constrained in a fully assembled Fusion 360 model with exploded view animation.

SKILLS USED

Autodesk Fusion 360 · Parametric Modeling · Bevel Gear Theory · Assembly Constraints · Engineering Visualization



Custom Differential Gearbox — Component Drawings

June 2025

PARAMETRIC DESIGN

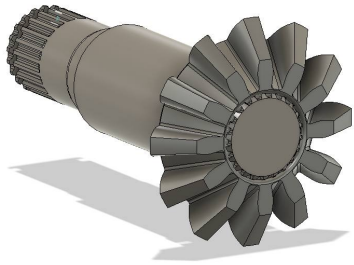
Every component dimension is driven by top-level gear parameters. Changing the module or tooth count propagates through the entire assembly without manual rework — a key advantage for iterative design.

KEY GEOMETRIES

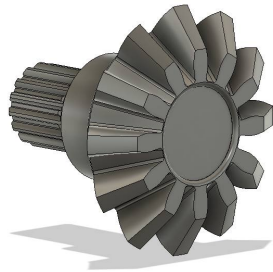
Bevel gears use a 20° pressure angle standard. The differential ring constrains the spider gear orbits and transmits input torque to the output shafts via the crown gear mesh.

VISUALIZATION

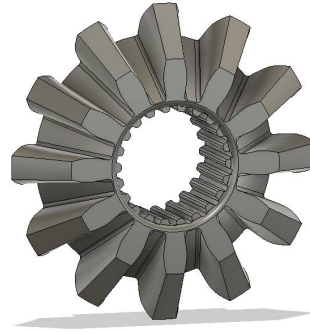
An exploded assembly animation was produced in Fusion 360 to communicate part relationships and assembly sequence clearly to non-engineering audiences.



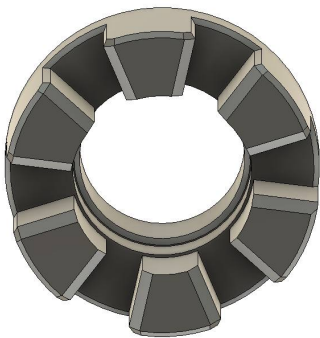
Long Drive Shaft



Short Drive Shaft



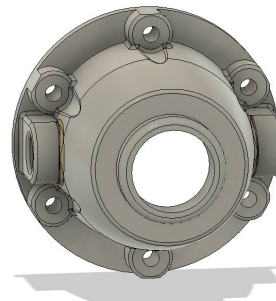
Bevel Gear



Differential Ring



Output Hub



Outer Casing



Self-Driving RC Truck with Obstacle Avoidance & Safety Filter

Mar – May 2025

OUTCOME

A 1/10-scale RC truck programmed in Python + ROS to complete two autonomous driving tasks. Task 1: navigated 16 waypoints while avoiding up to 20 randomly placed obstacle cubes within 3:30. Result: 4th fastest of 14 teams. Task 2: ADAS safety filter overrode steering commands that would cause track departure.

TASK 1 — OBSTACLE AVOIDANCE

Path planner dynamically resizes safety ellipse radii within a threshold distance of obstacles. Gaussian filtering ($\sigma=2$ physical, $\sigma=5$ sim) smoothed trajectories. Dual-planner cost calculation allowed simultaneous manual control until the last safe moment for full intervention.

TASK 2 — SAFETY FILTER

Two-layer safety system using real sensor measurements and a simulator running at 1000 Hz for soft trajectory generation. Costs calculated over a varying time horizon; partial intervention applied as late as safely possible.

KEY LESSON

Bridging the simulation-to-hardware gap required substantial parameter tuning. A boolean flag bug causing incorrect behavior across multiple scenarios was caught through systematic per-subsystem isolation testing.

2D Unsteady Heat Transfer: Heated Cylinder & NACA 2412 Airfoil

Apr – May 2025

OUTCOME

A Psi-Omega (streamfunction-vorticity) finite difference numerical simulation built from scratch in MATLAB. Two geometries studied: a 400°C cylinder in 300°C flow (convective heat flux 7.6 kW/m²), and a NACA 2412 airfoil at 5° and 30° AoA (heat flux 1,002.6 vs 1,133.6 W/m²). Each run took 3–6 hours.

TECHNICAL APPROACH

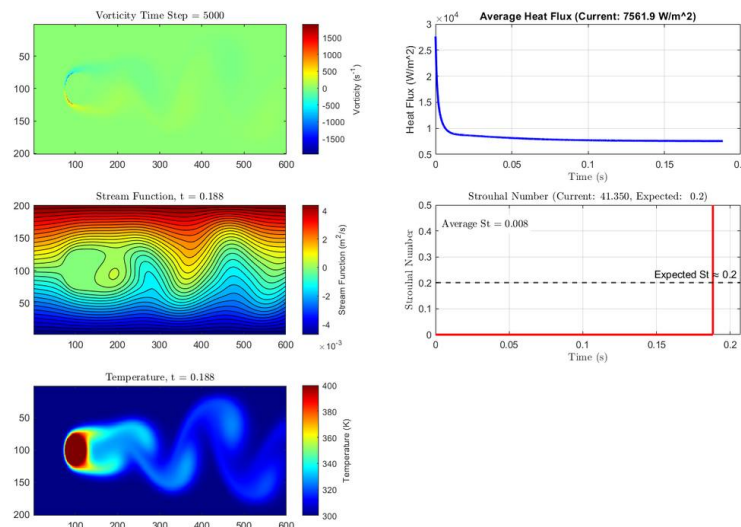
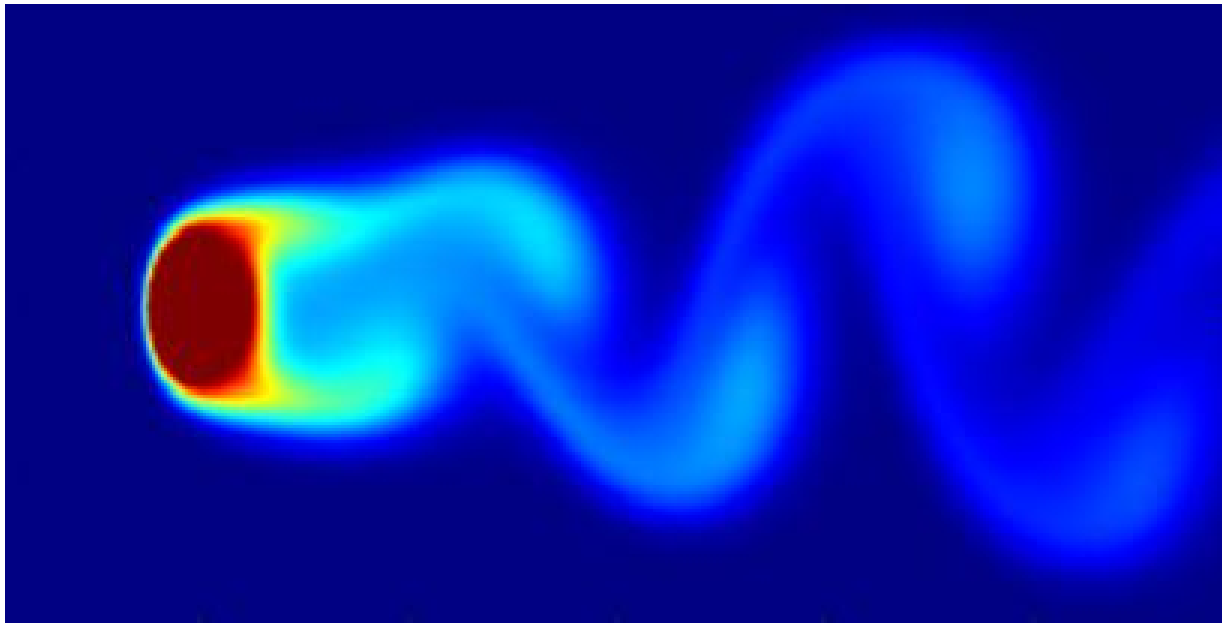
Three coupled PDEs solved at each time step: Streamfunction (Poisson) via Gauss-Seidel with over-relaxation; Vorticity (momentum) via forward Euler with upwind differencing; Temperature (energy) coupled to the velocity field.

VALIDATION

The 5° AoA result was validated against SimFlow CFD. The 30° AoA simulation exhibits vortex shedding with Strouhal number $St = 0.62$, consistent with separated flow theory.

SKILLS USED

MATLAB · Finite Difference Methods · Fluid Dynamics · Heat Transfer · Numerical Stability · CFD Validation



Heat Transfer Simulation — Airfoil Results

Apr – May 2025

5° ANGLE OF ATTACK

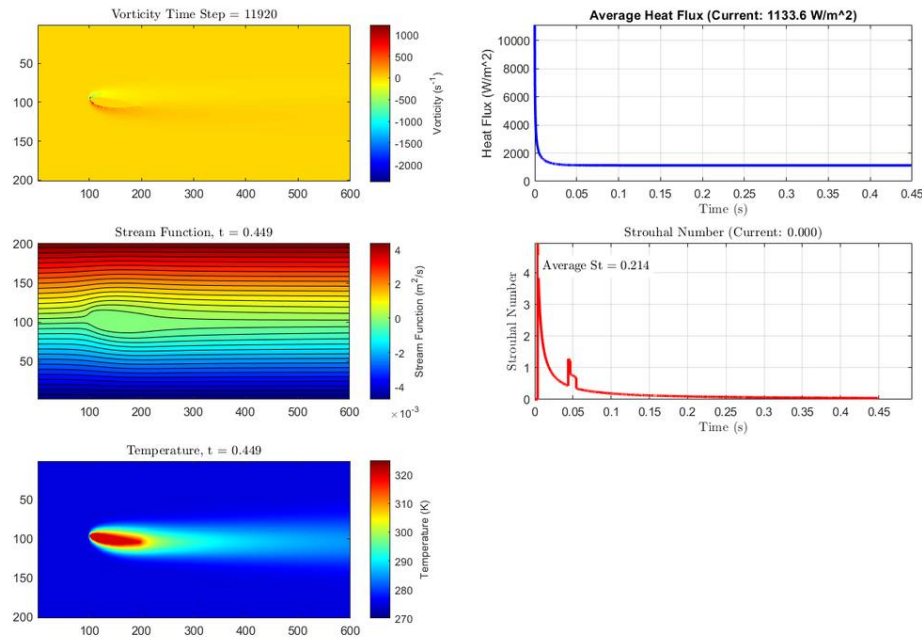
Flow remains attached. Temperature distribution is smooth and symmetric about the chord line. Validated within 5% of SimFlow commercial CFD results, confirming correct PDE discretization and boundary conditions.

30° ANGLE OF ATTACK

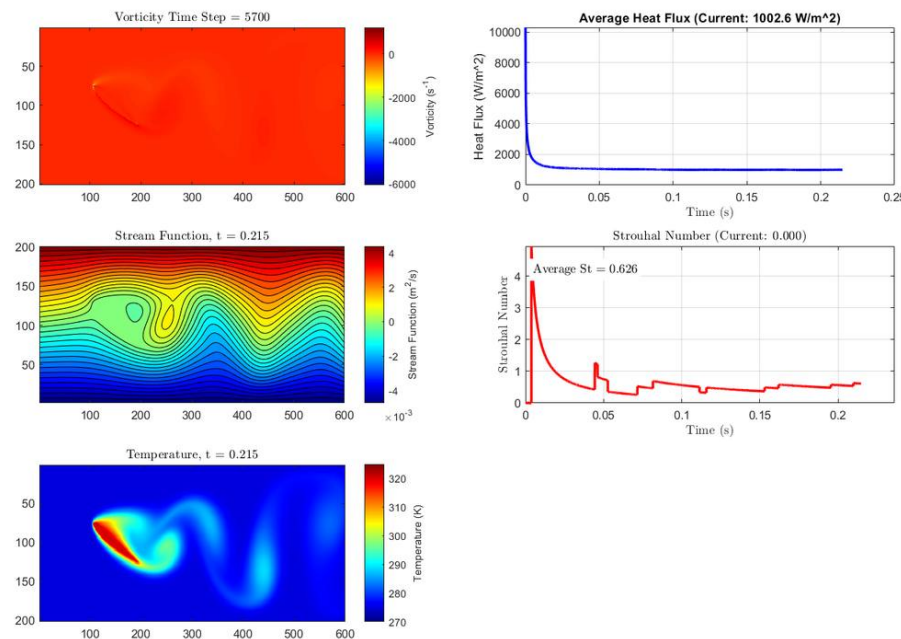
Flow separates at the leading edge, producing large unsteady vortices shed periodically into the wake. Higher stagnation temperatures and increased heat flux (13% above 5° case) result from thickened boundary layer.

DEBUGGING PROCESS

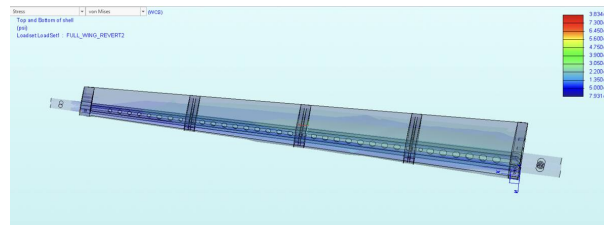
10 iteration cycles for the cylinder, 5 for the airfoil. Primary instability sources: vorticity boundary conditions at the cylinder wall and incorrect vorticity extrapolation at sharp airfoil edges.



NACA 2412 at 5° AoA — attached flow (validated against SimFlow)



NACA 2412 at 30° AoA — vortex shedding and stall ($St = 0.62$)



MECHANICAL ENGINEERING

Design and Manufacturing of a Load-Bearing Airfoil

Mar – May 2023

OUTCOME

A 9-member team designed and manufactured a 25.5-inch load-bearing airfoil that supported 97 lbf at 23.5 inches from the secured end with 3.75 in. tip deflection. Structure cost \$180.75, weighed 18.3 oz. Failure at 120 lbs (beam web torsion) — above the 97 lbf design load.

DESIGN

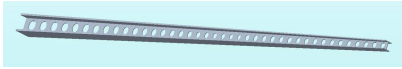
Aluminum 7075-T651 I-beam (yield stress 73,000 psi) with circular lightening holes selected for best strength-to-weight ratio from four competing beam concepts. Nylon bulkheads with circular cutouts reduced weight; balsa wood skin formed the NACA profile.

FEA & VALIDATION

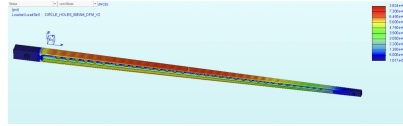
PTC Creo FEA validated deflection at 97 lbf. Failure mode — beam web torsion collapse at 120 lbs — was not captured in the planar FEA model, teaching the importance of 3D torsional load modeling.

SKILLS USED

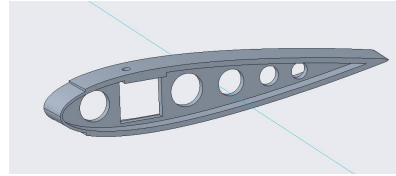
PTC Creo · FEA · CNC Milling (Haas VF-7/40) · Waterjet (ProtoMAX) · FDM & SLS 3D Printing · Structural Analysis · GD&T



Circular-Hole I-Beam



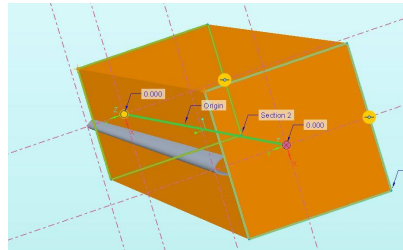
I-Beam FEA



Nylon Bulkhead Design



Waterjetted Bulkhead



Leading Edge



Pre-Test Assembly

MECHANICAL ENGINEERING

Load-Bearing Airfoil — Manufacturing & Testing

Mar – May 2023

MANUFACTURING

I-beam CNC milled on Haas VF-7/40. Bulkheads waterjet-cut on ProtoMAX. Ribs printed on Ender 3 Pro (FDM) and structural parts on Formlabs SLS. Balsa skin cut on bandsaw and sanded to NACA profile.

TESTING RESULTS

Passed 97 lbf design load at 3.75 in. deflection. Continued loading to failure at 120 lbs. All cost and weight targets met.

LESSONS LEARNED

Torsional loads must be explicitly modeled in 3D FEA. Physical load paths in complex assemblies differ from simplified beam theory predictions, especially near joints and web openings.